An Uncertainty-Aware Machine Learning and Multi-Objective Optimization Framework for Resilient Construction Material Procurement

Construction Material Procurement*

> *This is a draft manuscript assembled for supervisor review. It is built from frozen result artifacts and existing project documentation. It is not a submission-ready claim. Open research-side limitations are recorded in §12 and in the package README.*

Abstract

Construction-material procurement is made under several interacting uncertainties: material prices vary over time, material demand is uncertain, and supply-side behaviour including delivery performance and disruption can affect availability. This paper develops and evaluates an uncertainty-aware framework that links probabilistic price and demand forecasting with joint uncertainty propagation and multi-objective procurement optimization.

The framework has three stages. First, probabilistic machine-learning forecasts for construction-material price and demand are produced with calibrated predictive uncertainty rather than point predictions only. Second, the predictive demand uncertainty is propagated jointly with an admissible supply/process-duration representation to estimate shortage and service risk. Third, the resulting risk information is consumed by a multi-objective procurement optimization formulation producing cost, shortage and supply-risk trade-offs.

The empirical environment combines Singapore construction-material price and demand activity with the SUCCESS construction logistics benchmark used as an operational simulation environment. The controlled evaluation uses 11 locked forecast origins, 4 common materials, and a common realized realization path, with paired bootstrap inference and Benjamini–Hochberg multiplicity control at the forecast-origin level.

The central empirical question is whether retaining predictive uncertainty through the forecast, propagation, and decision layers produces measurably different procurement outcomes under the declared evaluation design, with the direction, magnitude, and desirability of those differences reported explicitly per objective, and with the cases where effects are statistically supported distinguished from the cases where they are not. The evidence is reported honestly across multiple objectives and configurations, including cases where effects are significant and cases where they are not. Scenario-based results are not presented as observed real-world outcomes, and no single controller is claimed universally optimal.

shortage and service risk; multi-objective optimization; resilience; reproducibility.

1. Introduction and research problem

Procurement of construction materials is exposed to several interacting sources of uncertainty. Prices for materials such as cement, aggregates, ready-mixed concrete and steel reinforcement evolve over time. Material demand on a project is uncertain. Supplier and logistics performance, including delivery timing and possible disruption, affects what can actually be supplied when it is needed. Procurement decisions — what to buy, when, in what quantity, and from which sources — affect cost, inventory, shortage, excess material, and the ability of a project to maintain its planned service and schedule.

A broad construction supply-chain and forecasting literature exists around price forecasting, demand forecasting, supply-chain risk, uncertain lead times, stochastic and robust procurement, supplier selection, and multi-objective optimization. This project does not claim novelty in any of those components individually. The narrower research problem is the **decision-centric handoff of predictive uncertainty**. Predictive information is not necessarily retained and propagated in a decision-active form from forecasting through

supply uncertainty and into procurement decisions. This project therefore asks whether predictive distributions, rather than only point forecasts, can produce measurably different procurement outcomes when transformed into shortage/service risk and consumed by a procurement decision model.

The central research question is:

> *Can predictive uncertainty be propagated through supply and process-duration uncertainty to produce measurably*
> *different construction-material procurement outcomes under a controlled evaluation design?*

The intended contribution is the controlled, uncertainty-preserving chain from forecasting to uncertainty to risk to decision, not a claim that forecasting, risk modelling, supplier selection, or optimization is individually novel.

2. Related work and evidence-supported research gaps

Related work in this area includes: construction-material price and demand forecasting; probabilistic, interval, and quantile prediction; construction supply-chain risk and disruption modelling; uncertain lead times and supplier reliability; stochastic, robust, and chance-constrained procurement and inventory formulations; supplier selection; and multi-objective procurement and cost–service trade-off analysis.

CMIDO does not treat any of those bodies of work as unobserved. The project's contribution is narrower and is framed around three gaps that are provisionally frozen from the current evidence audit and remain conditional on final formal literature screening:

exists (e.g. Assaf et al. 2024 doi:10.1061/9780784485286.026; Mir et al. 2021 doi:10.1016/j.jobe.2021.102288), the direct propagation of empirically calibrated predictive distributions into downstream construction-material procurement decisions is not fully established. The question is not whether probabilistic forecasting exists, but whether retaining predictive uncertainty produces measurably different procurement outcomes compared with equivalent point forecasts — and no verified screened study applies CQR or conformal prediction to Singapore material price/demand series with downstream decision evaluation.

model demand uncertainty and supply-side uncertainty in various forms (e.g. Chen et al. 2023 doi:10.1016/j.autcon.2023.104847; Dixit et al. 2014 doi:10.1016/j.cie.2013.12.009; Badkoubeh & Ghannadpour 2024 doi:10.3846/jcem.2024.21450), but the joint propagation into a decision-dependent shortage/service-risk distribution is not fully established. Shortage is treated primarily as a derived consequence of interacting uncertainties rather than as a separately trained classifier. No verified screened study propagates calibrated demand distributions through a joint demand–duration representation into a multi-objective procurement optimization with controlled ablation.

procurement decision value is not fully isolated through controlled component-level and decision-level comparisons. The question is how much of the final difference comes from forecasting uncertainty, formal optimization, and their integration. The screened evidence (Cluster B, §3 of the literature review) confirms that construction studies in the prior art compare policies or conservatism settings rather than factorial components; CMIDO's O1/O2/O3/O4 ablation is distinctive in its factorial isolation of optimization, predictive uncertainty, and their integration.

These gaps map to three research objectives: probabilistic forecasting, joint uncertainty propagation, and multi-objective procurement optimization.

3. Research objectives, questions, and contributions

construction-material price and demand, producing calibrated predictive distributions for downstream use.

RQ1: To what extent do probabilistic machine-learning forecasting models improve predictive accuracy, calibration, and uncertainty representation for construction-material price and demand compared with statistical and point-forecast baselines?

predictive demand uncertainty with an admissible supply/process-duration representation to quantify construction-material shortage and service risk.
RQ2: How does jointly propagating demand and supply/process-duration uncertainty affect shortage and service-risk estimates compared with deterministic, demand-only, and supply-only representations?
procurement optimization framework that converts predictive and supply-risk information into procurement quantity, timing, and supplier-allocation decisions.
RQ3: Does integrating predictive uncertainty and joint supply-risk information into procurement optimization produce measurably different cost–service–resilience trade-offs than deterministic and component-wise alternatives under normal and stressed conditions?
The intended contribution is the controlled forecast → uncertainty → risk → decision chain, evaluated with paired origins and explicit uncertainty and multiplicity handling, without claiming universal superiority.

4. Data sources, provenance, and admissible modelling constructs

Three data environments are used.
primary empirical RO1 environment. Prices are treated as observed quantities where the source supports that interpretation.
primary empirical RO1 demand environment.
simulation environment for RO2/RO3. It provides sites, suppliers, material demand, origin/destination structure, distances and travel times, and truck/capacity information. It is used for network structure, supplier/site relationships, operational demand, transport/logistics structure, and scenario/simulation behaviour. It is not treated as a complete historical procurement-order/delivery dataset.
This distinction matters: **SUCCESS travel time is not relabeled as historical procurement lead time**, and static supplier/catalog lead times are not represented as realized historical delivery lead times. Where a required construct is not observed, it is modeled as estimated or scenario-generated only, and that provenance is made explicit.

- **OBS** — directly observed/raw dataset quantity
- **DER** — deterministically derived from observed or frozen inputs
- **EST** — estimated or model-generated quantity
- **SCN** — scenario-generated quantity used for simulation/stress analysis

These labels are preserved in manuscript tables, figures, and supplementary material. OBS, DER, EST, and SCN are not silently interchanged.
representation. Its semantic role is preserved: process-duration data support the propagation construct they are documented for, and are not promoted to supplier-specific historical lead time without independent evidence.

5. RO1 probabilistic forecasting methodology and evaluation

RO1 develops probabilistic forecasts for construction-material price and demand. The forecasting role is to produce point predictions together with predictive uncertainty, so that downstream stages can consume distributions rather than only point estimates.
The candidate modelling space includes statistical baselines such as seasonal naive, ETS, and ARIMA/SARIMA where appropriate, and machine-learning baselines such as gradient boosting and XGBoost/LightGBM, with probabilistic or quantile extensions where supported. Deep or sequence models are used only where justified by the data and the forecasting role. Model selection is justified by literature, data characteristics, computational cost, forecasting role, and fairness of comparison rather than by maximizing the number of algorithms.
Forecast validation uses expanding-window rolling-origin backtesting rather than random train/test splits. The evaluation design uses multiple forecast horizons, with a primary tactical horizon unless the final application

analysis justifies another. Exact training/validation/test boundaries are generated from the final cleaned dataset rather than hard-coded.

Forecast evaluation separates **point accuracy** from **uncertainty quality**. Point metrics include MAE, RMSE, and sMAPE where appropriate. Probabilistic quality includes pinball loss, CRPS where supported, prediction-interval coverage, interval width/sharpness, and calibration. Where appropriate, forecast comparison uses Diebold–Mariano testing.

The manuscript does not conflate point accuracy with uncertainty quality. A model can be useful for uncertainty representation even when its point accuracy is not uniformly best, and vice versa. The conclusion reports both and does not equate non-significance with equivalence.

`results/forecasting/probabilistic_calibration/RO1_step26c3_validation_forecasts.csv`. The forecast evaluation metrics are reported in `results/8Q_publication_artifacts/T1_RO1_validation_metrics.csv` and reconciled against `results/forecasting/probabilistic_calibration/RO1_step26c3_validation_metrics.csv`.

6. RO2 joint uncertainty propagation and shortage/service-risk methodology

RO2 links predictive demand uncertainty with an admissible supply/process-duration representation to estimate shortage and service risk.

The conceptual mechanism is that demand during an uncertain lead/process duration is the accumulation of uncertain demand over an uncertain interval, which produces a distribution of required supply, an available-supply distribution, a shortage distribution, and a service probability. This is treated as a derived consequence of interacting uncertainties, not as a separately trained shortage classifier.

Propagation is implemented primarily through Monte Carlo methods, with documented convergence evidence for the Monte Carlo draw lock. Demand and process-duration dependence is not assumed independent automatically; dependence is tested where data permit.

The RO2 result families used in this manuscript are:

- joint propagation summary,
- tail comparison by representation,
- propagation summary across the evaluated population,
- service-risk curve,
- joint/independent sensitivity.

The frozen source result files are:

- `results/RO2/distribution/RO2_step27c3_joint_propagation_summary.csv`
- `results/RO2/distribution/RO2_step27c3_tail_comparison.csv`
- `results/RO2/propagation/RO2_step27d_propagation_summary.csv`
- `results/RO2/propagation/RO2_step27d_service_risk_curve.csv`
- `results/RO2/propagation/RO2_step27d_joint_vs_independent_sensitivity.csv`

The admissible process/duration representation is documented in

`results/RO2/data_audit/RO2_step27b4_admissible_modelling_view.csv`.

duration representation propagate into shortage/service-risk outcomes under the documented modelling view. It does not establish supplier-specific historical lead-time behaviour unless independently supported.

7. RO3 multi-objective procurement formulation, constraints, and evaluation

RO3 converts predictive and supply-risk information into procurement decisions through multi-objective optimization.

The decision vector includes procurement quantities, order activation, and safety stock. Supplier allocation is derived from procurement quantities. The formulation treats cost, shortage, and supply-risk exposure as

distinct objectives and produces a Pareto set rather than an arbitrarily weighted single optimum.
order quantity, storage capacity, and budget are considered. Feasibility is not asserted where constraint evidence is absent. Supplier allocation, safety stock, and storage/supplier capacity constraints that are not supported by registered evidence are explicitly unavailable rather than assumed zero.
the experiment reported here unless supported by registered evidence. The formulation notes that these variables are deliberately excluded from the decision surface in this experiment, not inferred from procurement quantities.

- O1 — deterministic information + heuristic control
- O2 — deterministic information + formal optimization
- O3 — probabilistic information + heuristic control
- O4 — probabilistic information + formal uncertainty-aware optimization

The common design uses 11 locked forecast origins, 4 common materials, and 12 realized months per origin/material, with common realized demand and price paths, an origin-level inferential unit, paired bootstrap uncertainty, and Benjamini–Hochberg multiplicity correction.
The controlled experiment is an attribution study. It can establish empirical differences among the predefined configurations, but it does not by itself establish broad causal or population-level superiority beyond the declared design.
`results/RO3/multi_origin/`, the controlled ablation sources in `results/RO3/ablation/final_comparison/`, and the stress/robustness artifacts in `results/RO3/ablation/RO3_7_stress/` and `results/RO3/ablation/RO3_8_robustness/`.
The publication tables used in this manuscript are:

- `results/8Q_publication_artifacts/T5_RO3_baseline_comparison.csv`
- `results/8Q_publication_artifacts/T5_T6_controller_descriptives.csv`
- `results/8Q_publication_artifacts/T6_RO3_ablation.csv`

These are reconciled against:

- `results/RO3/ablation/final_comparison/RO3_step36_statistical_pairwise_comparisons.csv`
- `results/RO3/ablation/final_comparison/RO3_step36_primary_ablation_results.csv`
- `results/RO3/ablation/final_comparison/RO3_step36_controller_descriptive_statistics.csv`

8. Experimental setup, baselines, ablations, statistical methods, and reproducibility

demand and price path across configurations. The inferential unit is the forecast origin, not individual monthly observations within an origin.
heuristic optimizers. O2 is a deterministic ablation of the formal optimization pipeline, not an independently developed optimization method. O4 is the integrated uncertainty-aware configuration.
Benjamini–Hochberg multiple-comparison control. Results report effect direction, uncertainty intervals, and significance status, without overstating population-level generalization from 11 repeated temporal cases.
records the environment, seeds where available, result paths, table/figure generation paths, and the exact regression command used for final verification.

9. Results

This section reports the frozen publication results. All numbers come from the 8Q publication tables and their declared source files. No new experiments are run in this packaging step.

9.1 RO1 forecasting and calibration

The RO1 evidence is reported from `results/8Q_publication_artifacts/T1_RO1_validation_metrics.csv`. The table reports dataset, series, horizon, sample sizes, point-accuracy metrics, and prequential coverage/width/Winkler results for the calibrated forecasts.

Key reporting discipline:

- Point accuracy and uncertainty quality are reported separately.
- Coverage and interval width are interpreted together, not in isolation.
- No claim is made that a forecasting model is universally best; performance is reported for the declared h series, and evaluation design.

The publication table is a calibrated-forecast-facing projection of the broader `results/forecasting/probabilistic_calibration/RO1_step26c3_validation_metrics.csv` source. It drops the `raw_*` coverage/width/Winkler columns and retains the prequential-calibrated columns; the shared numeric values are intact. The publication figure for this section is `results/8R_publication_figures/F1_RO1_calibration_coverage.png`, generated from `results/8Q_publication_artifacts/T1_RO1_validation_metrics.csv`.

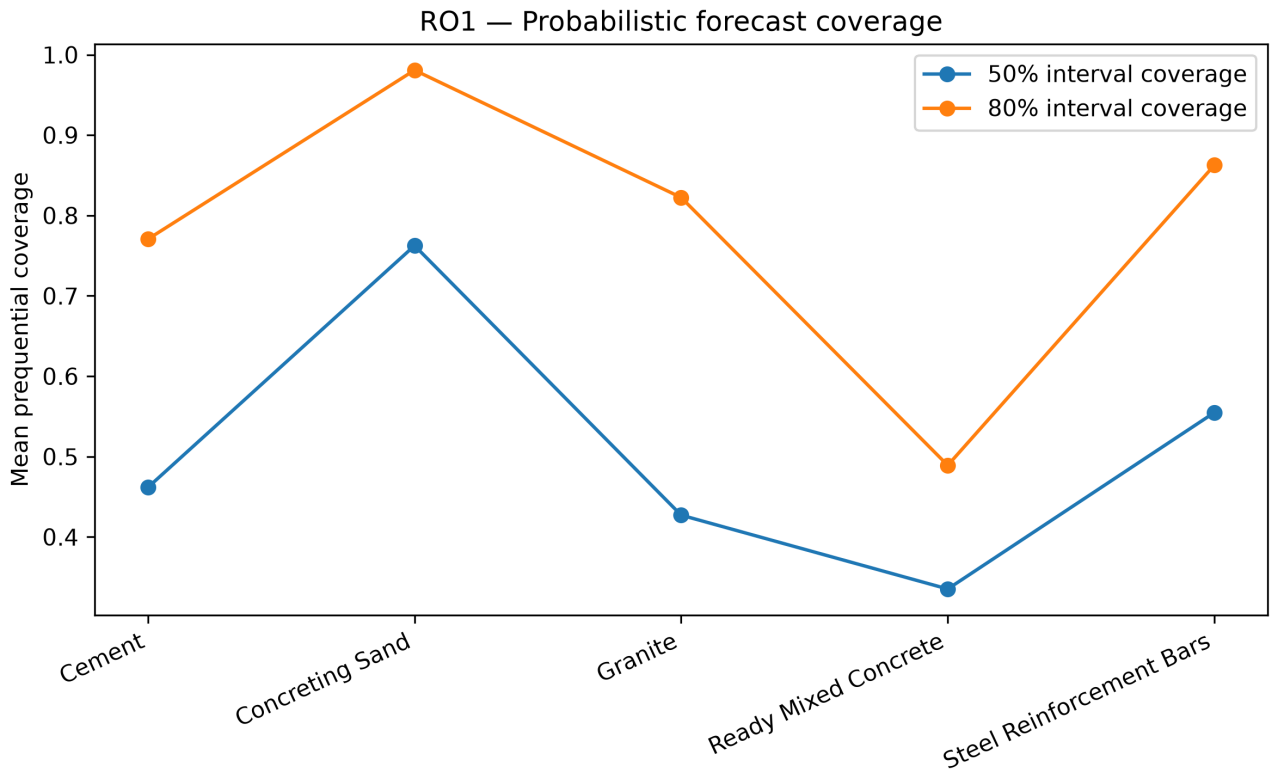


Figure 1 — RO1 / observational forecast calibration against observed failure counts

9.2 RO2 propagation and service risk

The RO2 evidence is reported from the 8Q publication tables derived from the RO2 source files listed in §6.

- `T2_RO2_joint_propagation_summary.csv` — joint propagation summary.
- `T2_RO2_tail_comparison.csv` — tail behaviour by representation.
- `T3_RO2_propagation_summary.csv` — propagated demand/shortage/service behaviour over the evalu
- `T3_RO2_service_risk_curve.csv` — service-risk curve.
- `T3_RO2_joint_independent_sensitivity.csv` — sensitivity to joint versus independent propagation assu

The manuscript does not claim that the admissible duration representation equals supplier-specific historical lead time. It reports the propagation behaviour under the documented admissible representation and notes the sensitivity of conclusions to the joint/independent assumption.

The publication figures for this section are `results/8R_publication_figures/F2_RO2_tail_comparison.png` (generated from `results/8Q_publication_artifacts/T2_RO2_tail_comparison.csv`), `results/8R_publication_figures/F3_RO2_propagation_tail.png` (generated from `results/8Q_publication_artifacts/T3_RO2_propagation_summary.csv`), and

`results/8R_publication_figures/F4_RO2_joint_independent_sensitivity.png` (generated from `results/8Q_publication_artifacts/T3_RO2_joint_independent_sensitivity.csv`).

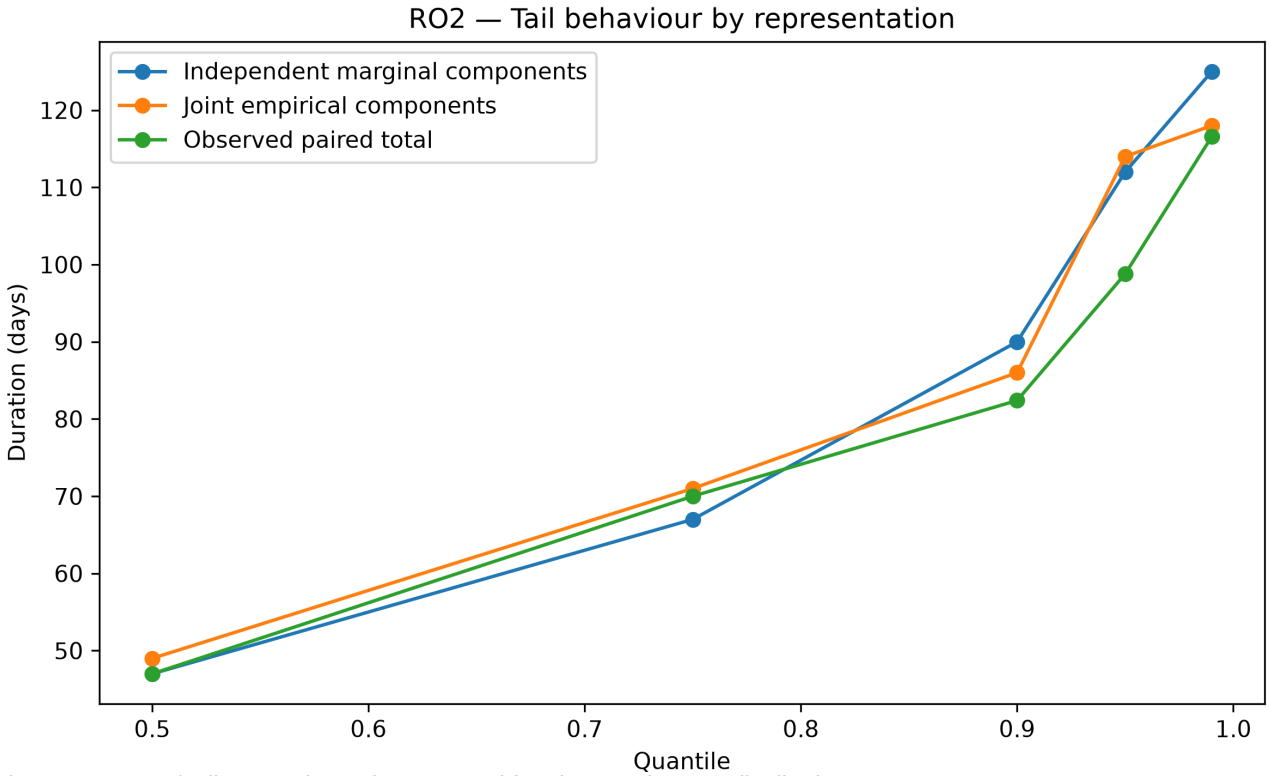


Figure 2 — RO2 / tail comparison of propagated frontier vs reference distributions

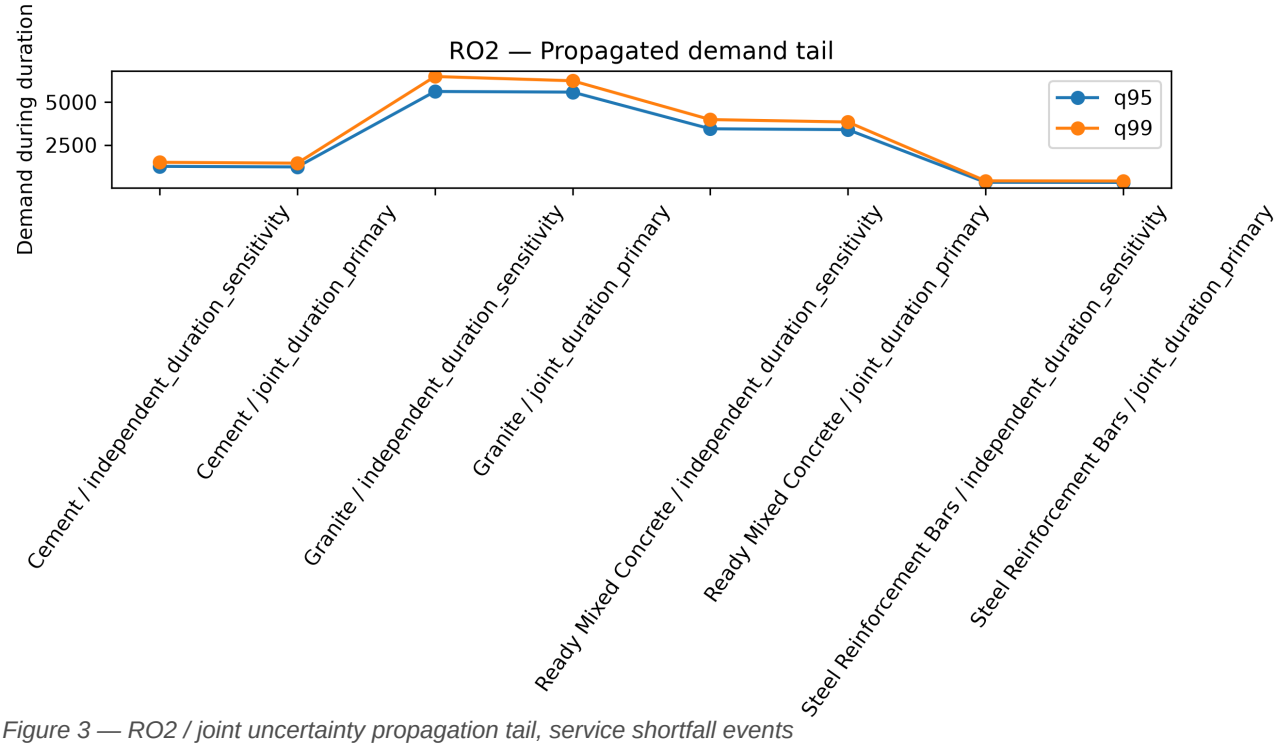


Figure 3 — RO2 / joint uncertainty propagation tail, service shortfall events

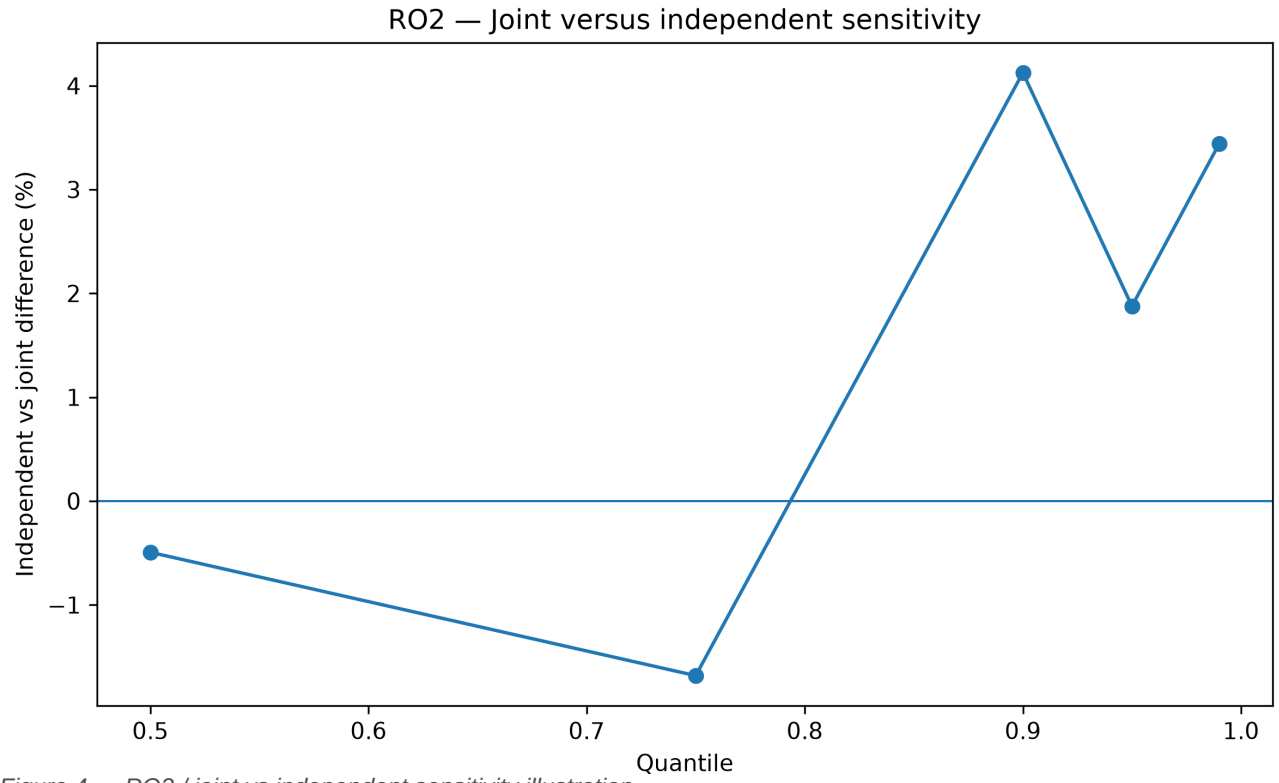


Figure 4 — RO2 / joint vs independent sensitivity illustration

9.3 RO3 baseline comparison and controller descriptives

The RO3 controlled evidence is reported from:

- `results/8Q_publication_artifacts/T5_RO3_baseline_comparison.csv`
- `results/8Q_publication_artifacts/T5_T6_controller_descriptives.csv`

The controller descriptive table reports, for each controller, the realized holding cost, shortage, service level, shortage CVaR, procurement quantity, ending inventory, and procurement-event summaries across the 11 locked origins.

The baseline comparison table reports the planned contrasts among O1, O2, O3, and O4. The contrasts reported include:

- O2 versus O1 — optimization under deterministic information
- O3 versus O1 — probabilistic information under heuristic control
- O4 versus O2 — incremental uncertainty under optimization
- O4 versus O3 — incremental optimization under probabilistic information
- O4 versus O1 — integrated O4 versus deterministic heuristic

The reported contrasts distinguish **differences in one objective** from **trade-offs in another**. For example, the O2 versus O1 contrast can show lower holding cost with higher shortage, while the O4 contrasts show materially different cost/shortage/service trade-offs from the deterministic and probabilistic-heuristic configurations.

The publication baseline table is a projection of the source `results/RO3/ablation/final_comparison/RO3_step36_statistical_pairwise_comparisons.csv`. It keeps the core contrast/metric/estimate/inference columns and drops additional per-contrast control columns; the shared numeric values are identical. The publication controller-descriptive table is generated from `results/RO3/ablation/final_comparison/RO3_step36_controller_descriptive_statistics.csv`. The publication figure for this section is `results/8R_publication_figures/F5_RO3_controller_descriptives.png`, generated from `results/8Q_publication_artifacts/T5_T6_controller_descriptives.csv`.

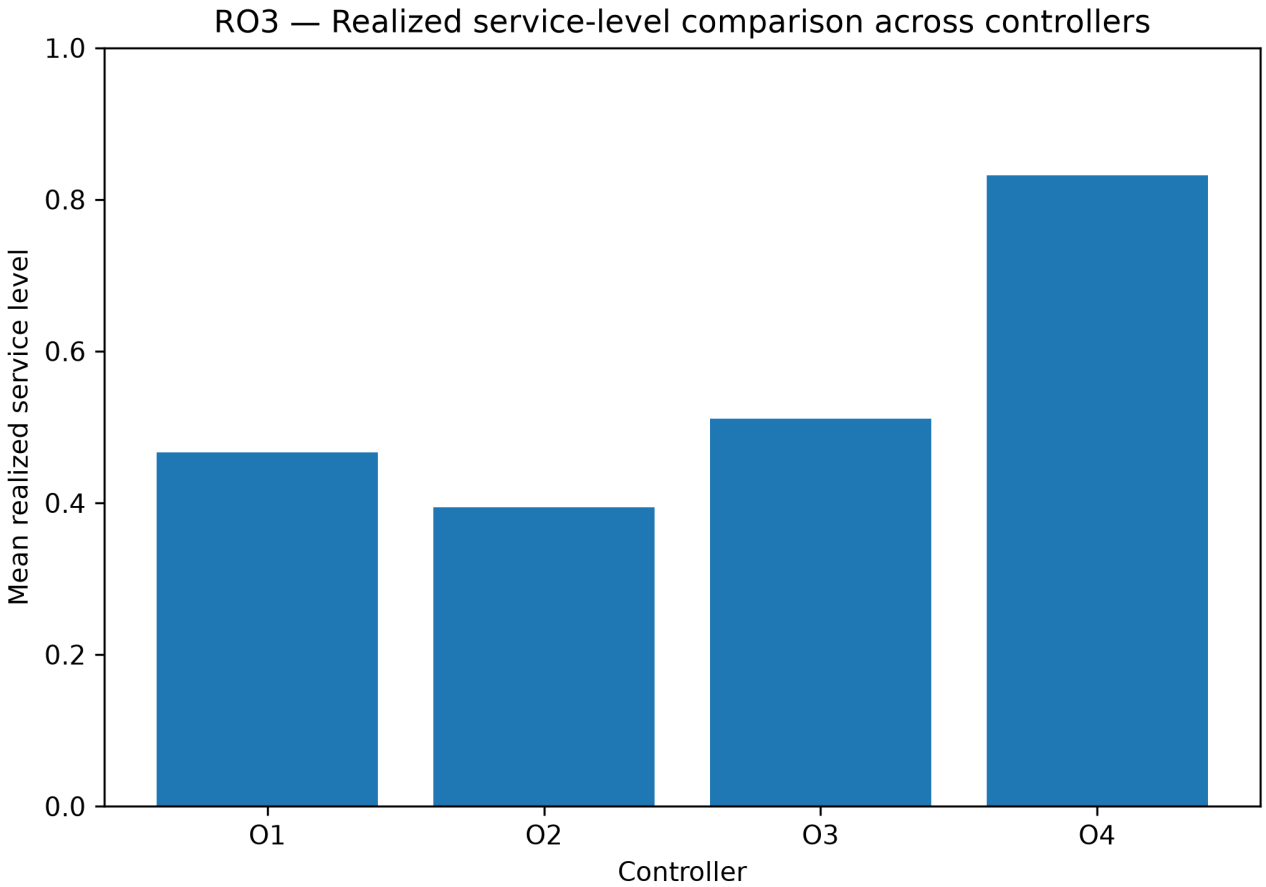


Figure 5 — RO3 / controller ensemble descriptives (means with variabilities)

9.4 RO3 ablation

The RO3 ablation evidence is reported from `results/8Q_publication_artifacts/T6_RO3_ablation.csv`. The ablation table reports the same contrasts as the baseline table but with the richer per-contrast detail needed for the ablation interpretation: relative change, bootstrap bounds, improved/worsened/tie counts, rank-biserial sign, and significance after multiplicity correction.

The ablation interpretation emphasizes attribution: which component changes are associated with which objective changes, and where significance is and is not achieved. The manuscript does not treat non-significance as evidence of equivalence.

The ablation table is generated from `results/RO3/ablation/final_comparison/RO3_step36_primary_ablation_results.csv` and shares its schema and values. The publication figure for this section is `results/8R_publication_figures/F6_RO3_ablation_effects.png`, generated from `results/8Q_publication_artifacts/T6_RO3_ablation.csv`.

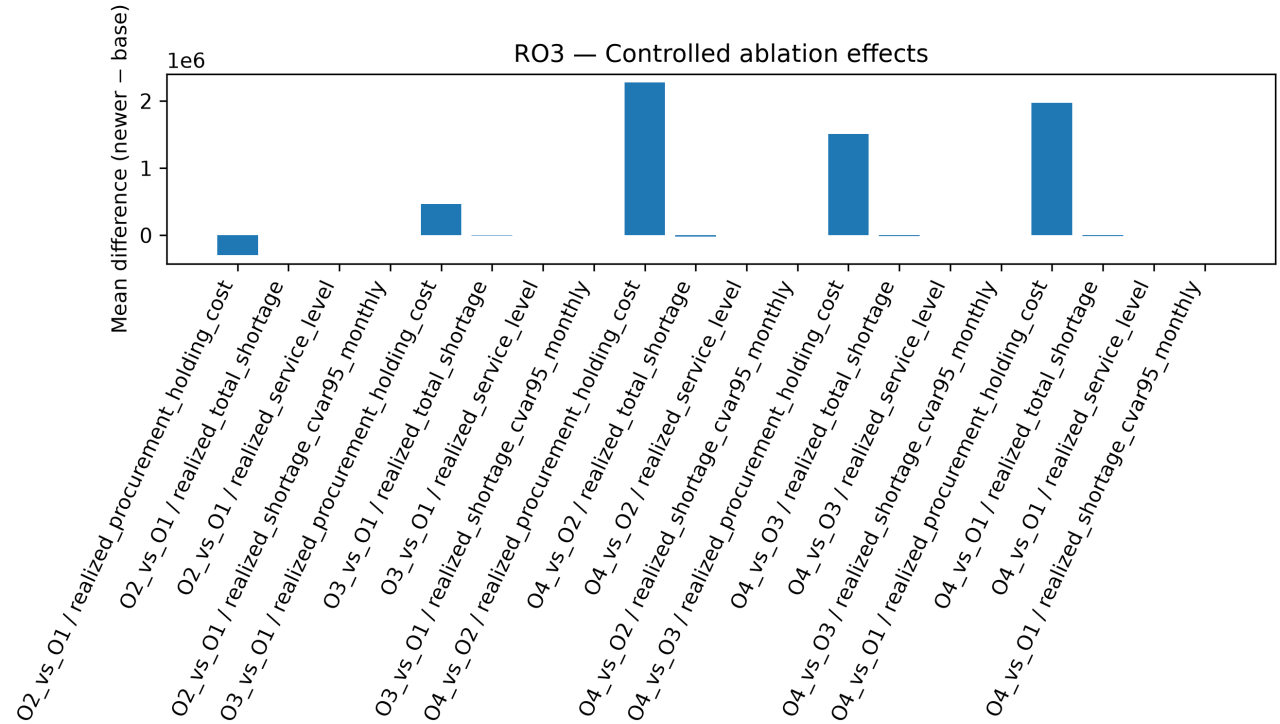


Figure 6 — RO3 / ablation/robustness sensitivity effects across controllers

9.5 Stress and robustness

Stress and robustness results are reported as manuscript-bound evidence in Table T4 (`results/8Q_publication_artifacts/T4_RO3_stress_robustness.csv`, packaged as `03_TABLES/T4_RO3_stress_robustness.csv`). T4 is generated by `src/optimization/cmido_8q_publication_artifacts.py` as a relabel-and-concatenate projection of two frozen upstream summaries — no value is recomputed — and every row is reconciled against both upstream files by `src/optimization/cmido_8s_numerical_reconciliation.py` (28 rows: 16 duration-stress + 12 holding-rate; both upstream SHA-256 hashes are recorded in the 8Q manifest).

Upstream stress artifacts:

- `results/RO3/ablation/RO3_7_stress/RO3_step37_stress_summary.csv`
- `results/RO3/ablation/RO3_7_stress/RO3_step37_stress_paired_changes.csv`
- `results/RO3/ablation/RO3_7_stress/RO3_step37_stress_audit.csv`

Upstream robustness artifacts:

- `results/RO3/ablation/RO3_8_robustness/RO3_step38_holding_rate_summary.csv`
- `results/RO3/ablation/RO3_8_robustness/RO3_step38_holding_rate_controller_contrasts.csv`
- `results/RO3/ablation/RO3_8_robustness/RO3_step38_holding_rate_sensitivity_vs_H30.csv`
- `results/RO3/ablation/RO3_8_robustness/RO3_step38_holding_rate_audit.csv`

What T4 shows, within the declared design: under the duration-stress ladder (P50 → P95), mean service level falls for every controller (O1 0.836 → 0.671, O2 0.613 → 0.443, O3 0.840 → 0.677, O4 0.832 → 0.669) and mean shortage roughly doubles (e.g. O1 6,183 → 12,384). Under the holding-rate ladder (H20 → H40), mean cost rises for every controller (e.g. O1 1.92M → 2.25M, O4 3.19M → 3.94M) while mean shortage and mean service are unchanged across holding cases in this design — the holding rate acts through the cost objective only.

These results report variation under declared stress and holding-rate sensitivity settings; they are not a claim that all procurement behaviour under all regimes is established, and no stress/robustness figure is bound to this section in this draft.

9.6 Integrated decision-value summary

The integrated claim reports whether the complete forecast → uncertainty → risk → decision chain is supported by the combined evidence set, with explicit scope. The current evidence supports reporting that the chain is empirically represented in the frozen artifacts — specifically `results/8Q_publication_artifacts/` (T1, T2, T3, T5, T5_T6, T6) and `results/8R_publication_figures/` (F1, F2, F3, F4, F5, F6) — and that the controlled contrasts show measurable differences among the predefined configurations. The underlying RO1/RO2/RO3 source files are listed in the review copy source map. The manuscript does not claim universal superiority or claim that the result generalizes beyond the declared evaluation design without additional evidence.

The implemented pipeline that produces, propagates and validates this chain is rendered as Figure 7 (`02_FIGURES/F7_CMIDO_integrated_pipeline.png`), a conceptual diagram generated by `src/optimization/cmido_8t_f7_pipeline_figure.py`. The diagram encodes the provenance class of each stage, contains no data-derived quantities, and carries the standing caveat that supplier-specific lead-time evidence is not inferred where it is unavailable.

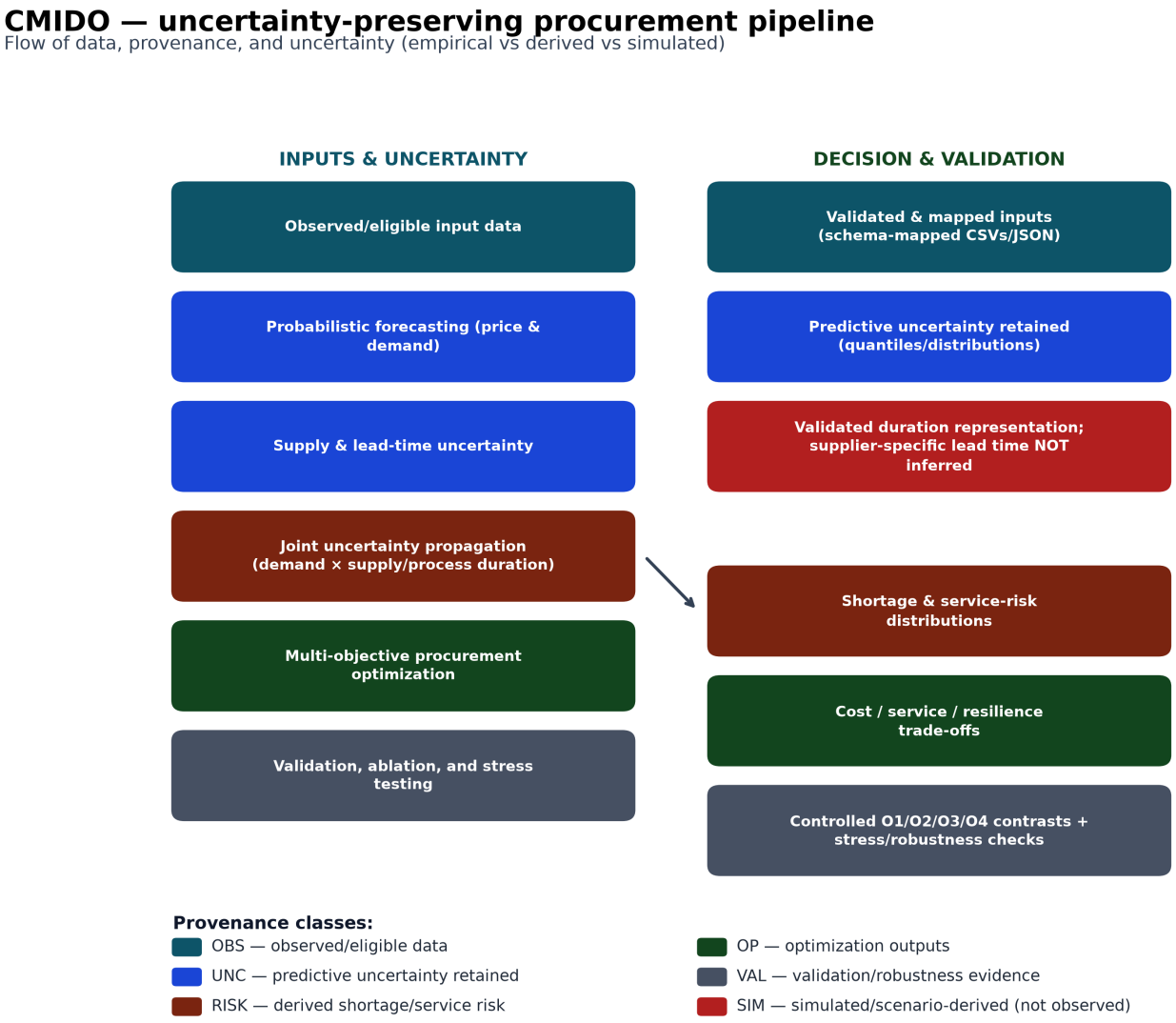


Figure 7 — CMIDO integrated pipeline: empirical inputs -> forecasts -> uncertainty -> joint simulation -> optimization -> validation.

10. Discussion and interpretation of incremental decision value

The discussion interprets the results as an attribution study.

The O1 → O2 contrast isolates the effect of formal optimization under deterministic information. The O1 → O3 contrast isolates the effect of probabilistic information under heuristic control. The O2 → O4 and O3 → O4 contrasts isolate the incremental effect of integrating uncertainty-aware information with optimization. The O1 → O4 contrast reports the integrated effect against the deterministic heuristic baseline.

Interpretation discipline:

- A contrast that differs in one objective may differ in another; this is reported as a trade-off, not as unambiguous improvement.
- Significance is reported per contrast and per metric with multiplicity control.
- Non-significant results are reported honestly and not treated as equivalence.
- The O4 configuration does not dominate every objective in every contrast; the result is a Pareto-relevant

trade-offs rather than a single winner.

- The controlled experiment uses 11 origins; generalization claims are therefore limited and explicitly scop

The framing of incremental decision value is therefore: under the declared evaluation design, the uncertainty-aware configuration changes the cost, shortage, and service trade-off surface relative to the deterministic and probabilistic-heuristic configurations in ways that are measurable and, for several contrasts, statistically supported. The magnitude and desirability of those trade-offs depend on the objective weighting and on the evaluation scope, which are not claimed to be universal.

11. Limitations, assumptions, and threats to validity

Assumptions requiring explicit disclosure

1. The selected RO1 predictive representation is treated as the frozen input to downstream decision experiments. 2. The RO2 admissible duration representation is used according to its documented semantic role; it is not relabeled as supplier-specific historical lead time unless independently supported. 3. Scenario-generated uncertainty is distinguished from direct observation. 4. The RO3.6 comparison uses paired forecast origins rather than treating monthly observations within an origin as independent replicates. 5. O1 and O3 are intentionally transparent heuristic benchmarks, not state-of-the-art heuristic optimizers. 6. O2 is a deterministic ablation of the formal optimization pipeline, not an independently developed optimization method. 7. O4 realized-policy evaluation and O4 stochastic optimization objectives are distinct analytical layers and are not conflated. 8. Any final manuscript aggregation preserves the exact evaluation population and frozen controller/configuration.

Threats to validity

- **Construct validity.** Forecast uncertainty, process duration, shortage, service, and resilience are repre through operational variables and model constructs; their correspondence to real procurement phenomena must be documented rather than assumed.

- **Internal validity.** Temporal leakage, outcome-based tuning, inconsistent evaluation populations, and origin removal would undermine attribution. The frozen controller and common-realization gates are designed to constrain these risks.

- **Statistical validity.** 11 origins are repeated temporal cases; small-sample paired inference should be r with uncertainty intervals and multiple-comparison control, without overstating population-level generalization.

- **External validity.** The primary empirical environment is Singapore construction-material activity and th SUCCESS construction logistics benchmark. Results should not be presented as automatically representative of all geographies, contractors, suppliers, materials, or market regimes.

- **Implementation validity.** Result-producing scripts should remain executable from a clean checkout; h developer-specific absolute paths are treated as reproducibility defects.

Explicit data limitations

- SUCCESS travel time is not relabeled as historical procurement lead time.
- Static supplier/catalog lead times are not represented as realized historical delivery lead times.
- Supplier capacity, minimum order quantities, storage capacity, and disruption/partial-delivery probabilitie treated as estimated or scenario-generated where not observed, and their provenance is kept explicit.

Package-level limitations

1. Five decision-level artifacts are registered but not present in the artifact inventory: `RO3_O2_SELECTED_DECISIONS`, `RO3_O2_REALIZED_EVALUATION_SUMMARY`, `RO3_O4_REALIZED_EVALUATION_LEDGER`, `RO3_MULTI_ORIGIN_SUMMARY`, `RO3_DECISION_SENSITIVITY`. 2. `RO1_PRIMARY_H3_SUMMARY` and `EVIDENCE_8P_MANIFEST` are registered but not adapted into evidence bundles. 3. Reported SHA-256 fingerprints have not been

recomputed and verified in this packaging session. 4. Presence checks for unregistered files are existence-only. 5. RO3 500→1,000 scenario convergence failed; the recorded next required evaluation is 2,500, then 5,000. 6. Observed-stream dates, authorship, and software-version metadata are unavailable. 7. Additional master-spec stress scenarios have no recorded results; none were fabricated. 8. Simulated/derived/scenario results are never described as observed real-world outcomes.

12. Conclusion and future work

This paper developed and evaluated an uncertainty-aware forecast → propagation → risk → decision framework for construction-material procurement. The framework links probabilistic price and demand forecasting, joint uncertainty propagation with an admissible supply/process-duration representation, and multi-objective procurement optimization.

Under the declared evaluation design, the evidence supports reporting measurable differences among the predefined O1/O2/O3/O4 configurations and explicit trade-offs among cost, shortage, and service objectives. The integrated uncertainty-aware configuration changes the procurement trade-off surface relative to deterministic and probabilistic-heuristic configurations in ways that are measurable and, for several contrasts, statistically supported. The work does not claim universal superiority, does not relabel travel-time data as historical lead time, and does not present scenario results as observed real-world outcomes.

Future work should include:

- resolving the remaining decision-level evidence registration and adaptation gaps;
- completing the RO3 scenario-convergence evaluation path specified for the failed 500→1,000 gate;
- establishing supplier-specific historical lead-time evidence only where independently supported;
- broadening the empirical environments if broader claims are to be made;
- completing the final manuscript claim-to-evidence concordance and a rendered review copy;
- freezing the final repository state and reproducibility package.

13. References and supplementary-material index

References are intentionally not invented in this draft. Any citation used in the final manuscript must support the statement it is attached to. The final reference list will be assembled from the actual literature used during the research and will be checked against the claims it supports. The verified starting set from the 2026-10 literature screening is provided below; a formal systematic literature review (SLR) per master spec §37 remains required for submission.

#	Citation	DOI / URL	Establishes
A1	Mir, Kabir, Nasirzadeh & Khosravi (2021), *Journal of Building Engineering* — "Neural network-based interval forecasting of construction material prices"	[doi:10.1016/j.jobe.2021.102288](https://doi.org/10.1016/j.jobe.2021.102288)	LUBE interval forecasts for US asphalt/steel; intervals outperform point estimates on their metrics
A2	Assaf, Assaad, El-adaway & Nabi (2024), *Construction Research Congress 2024* — DeepAR probabilistic construction-cost forecasting	[doi:10.1061/9780784485286.026](https://doi.org/10.1061/9780784485286.026)	DeepAR Monte-Carlo price samples for four material categories (conference paper)
A3	Jezzini, Assaad & El-Adaway (2025), *J. Construction Engineering and Management* 151(7) — probabilistic deep learning → project cost-risk from material price volatility	[doi:10.1061/JCEMD4.COENG-16055](https://doi.org/10.1061/JCEMD4.COENG-16055)	Predictive price distributions feed stochastic contract price-escalation risk analysis
A4	Fouad, Wefki & Kouta (2025), *J. Supply Chain Management Science* — AI-driven raw-material demand forecasting	[doi:10.59490/jscms.2025.8136](https://doi.org/10.59490/jscms.2025.8136)	XGBoost/LSTM/RF demand models; reported 66.5% overstock reduction in planning
A5	Ma, Tam, Le & Osei-Kyei (2024), *Int. J. Construction Management* 24(11) — SLR on price-forecasting models in construction		

[doi:10.1080/15623599.2023.2241761](https://doi.org/10.1080/15623599.2023.2241761) | 34 studies 2012–2022; VECM/ANN/ARIMA dominate; probabilistic calibration not a recurring practice |
| # | Citation | DOI / URL | Establishes | |---|----------|-----------|-------------| | B1 | Chen et al. (2023), *Automation in Construction* 150 — supplier selection and material purchasing for the construction supply chain in a fuzzy scenario-based environment | [doi:10.1016/j.autcon.2023.104847](https://doi.org/10.1016/j.autcon.2023.104847) | Multi-objective MILP integrating supplier selection, inventory management, order quantities, split orders; prices, capacities, delays as fuzzy parameters; >55% cost increase with greater price variation, >70% weighted lateness with greater delay variation | | B2 | Presbitero, Syrén, Dippel, Awasthi, Dev & Nie (2025), *Scientific Reports* — replenishment optimization with extended (r,s,Q) and probabilistic models | [doi:10.1038/s41598-025-32537-2](https://doi.org/10.1038/s41598-025-32537-2) | Conformal-prediction quantiles consumed by simulation-assisted replenishment optimizer with ablations; e-commerce, not construction | | B3 | Dixit, Srivastava & Chaudhuri (2014), *Computers & Industrial Engineering* 76 — procurement scheduling with fuzzy activity durations and lead times | [doi:10.1016/j.cie.2013.12.009](https://doi.org/10.1016/j.cie.2013.12.009) | Fuzzy activity durations and lead times in procurement scheduling; reported fuzzy holding and shortage costs | | B4 | Badkoubeh & Ghannadpour (2024), *J. Civil Engineering and Management* 30(7) — construction supply chain model with backup supplier for resilience | [doi:10.3846/jcem.2024.21450](https://doi.org/10.3846/jcem.2024.21450) | Scenario-based MIP with primary and backup suppliers; minimizes logistics costs and delays under disruption scenarios |
| # | Citation | DOI / URL | Note | |---|----------|-----------|------| | C1 | JCEM 2025 — material-price CatBoost model | [doi:10.1061/JCEMD4.COENG-16748](https://doi.org/10.1061/JCEMD4.COENG-16748) | Recent peer-reviewed precedent in target journal | | C2 | JCEM 2025 — cost-uncertainty detection | [doi:10.1061/JCEMD4.COENG-15689](https://doi.org/10.1061/JCEMD4.COENG-15689) | Recent peer-reviewed precedent in target journal | | C3 | Automation in Construction 2024 — BIM/blockchain procurement system | [doi:10.1016/j.autcon.2024.105779](https://doi.org/10.1016/j.autcon.2024.105779) | Recent peer-reviewed precedent in target journal | | C4 | ECAM 2025 — ML cost-overrun prediction | [doi:10.1108/ECAM-09-2024-1209](https://doi.org/10.1108/ECAM-09-2024-1209) | Recent peer-reviewed precedent in target journal | | C5 | IJPE 2024 — multi-channel procurement under disruption | [doi:10.1016/j.ijpe.2024.109346](https://doi.org/10.1016/j.ijpe.2024.109346) | Recent peer-reviewed precedent in target journal |

- the frozen publication tables in `results/8Q_publication_artifacts/`;
- the frozen publication figures in `results/8R_publication_figures/`;
- the upstream RO1/RO2/RO3 source files listed above;
- the 8M/8N/8O/8P/8Q/8R/8S manifests and audit files;
- the reproducibility package in `06_REPRODUCIBILITY/` of this folder.

14. Data/code availability and reproducibility statement

The frozen result files used in this manuscript are stored in the CMIDO repository under `results/`. The publication tables and figures are generated from those sources by the scripts:

- `src/optimization/cmido_8q_publication_artifacts.py`
- `src/optimization/cmido_8r_publication_figures.py`
- `src/optimization/cmido_8s_numerical_reconciliation.py`

The final regression suite used for verification is:

[code block]

```
cd /d/CMIDO
```

```
python -m pytest tests/construction/ tests/optimization/ -q --tb=line
```

The current full regression result for this packaging session was 1059 passed with exit code 0.
The final package manifest is recorded in `06_REPRODUCIBILITY/` and in `results/CMIDO_FINAL_PACKAGE_MANIFEST.json`. It records the repository commit, environment summary, and the status of each evidence gate.
manifest-defined checks and the numeric reconciliation in 8S. It does not claim that every result is externally validated, causally established, or transferable beyond the declared evaluation design. Readers should treat scenario, estimated, and derived quantities according to their provenance labels.

Change log / conditional items

- Tables and figures: frozen 8Q/8R artifacts, reconciled in 8S. No new scientific results generated in this packaging session.
- Manuscript: draft assembled from existing evidence package and frozen artifacts. Numerical claims are t the 8Q tables and their declared source files.
- Wording fixes applied in this packaging step:
- the central empirical claim was tightened so that the manuscript reports *measurably different procureme outcomes with explicit direction, magnitude, desirability, and significance status per objective*;
- the abstract no longer implies a directional overall superiority claim; it reports *measurably different outcomes* and explicitly distinguishes statistically supported effects from non-significant ones.
- Remaining open items: see §11 and the package README.

<!-- End of editable draft. Review copy is separate. -->

CMIDO_DRAFT_MANUSCRIPT.pdf — rendered review copy